

Name _____

Section 1-3 - Using Midpoint and Distance Formulas

With Your Table Partner, Answer as Many as You Can:

1. Draw a number line 0 to 6. What is the midpoint?



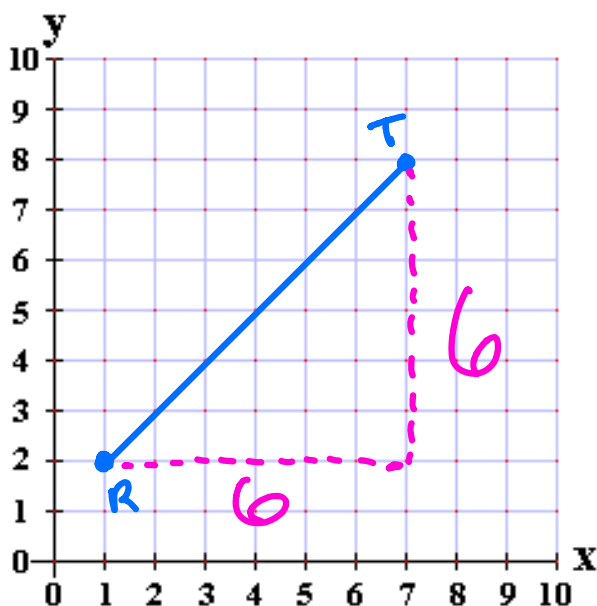
3

2. If you score 70% on a test and 94% on another test, what is your average test score?

$$\frac{(70 + 94)}{2}$$

82%

3. On the graph, plot $R(1, 2)$ and $T(7, 8)$. What is the midpoint of $\overline{RT}$?



$M(4, 5)$

4. What is the length of $\overline{RT}$?

$$6^2 + 6^2 = c^2$$

$$c^2 = 72$$

$$\sqrt{72}$$

$$6\sqrt{2}$$

≈ 8.5 units

Midpoint Formula

With endpoints $A(x_1, y_1)$ and $B(x_2, y_2)$, M is the midpoint of $\overline{AB}$.

$$(4, 5)$$

$$M\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$$

$$\left(\frac{(1+7)}{2}, \frac{(2+8)}{2}\right)$$

Example:

Find the coordinates of the midpoint M if the endpoints of $\overline{KT}$ are

$K(-42, 67)$ and $T(88, -76)$.

x_1, y_1

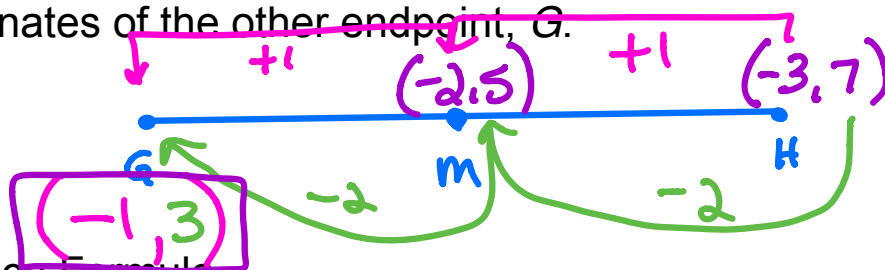
x_2, y_2

$$\left(\frac{(-42+88)}{2}, \frac{(67+(-76))}{2}\right)$$

Example:

$$M(23, -\frac{9}{2}) \leftarrow \text{or } -4.5$$

The midpoint of $\overline{GH}$ is $M(-2, 5)$. One endpoint is $H(-3, 7)$. Find the coordinates of the other endpoint, G .



$$\frac{(-3+x)}{2} = -2$$

$$\frac{(7+y)}{2} = 5$$

$$7+y=10$$

Distance Formula

In a graph, use the Pythagorean Theorem $a^2 + b^2 = c^2$ OR the Distance Formula with $A(x_1, y_1)$ and $B(x_2, y_2)$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example:

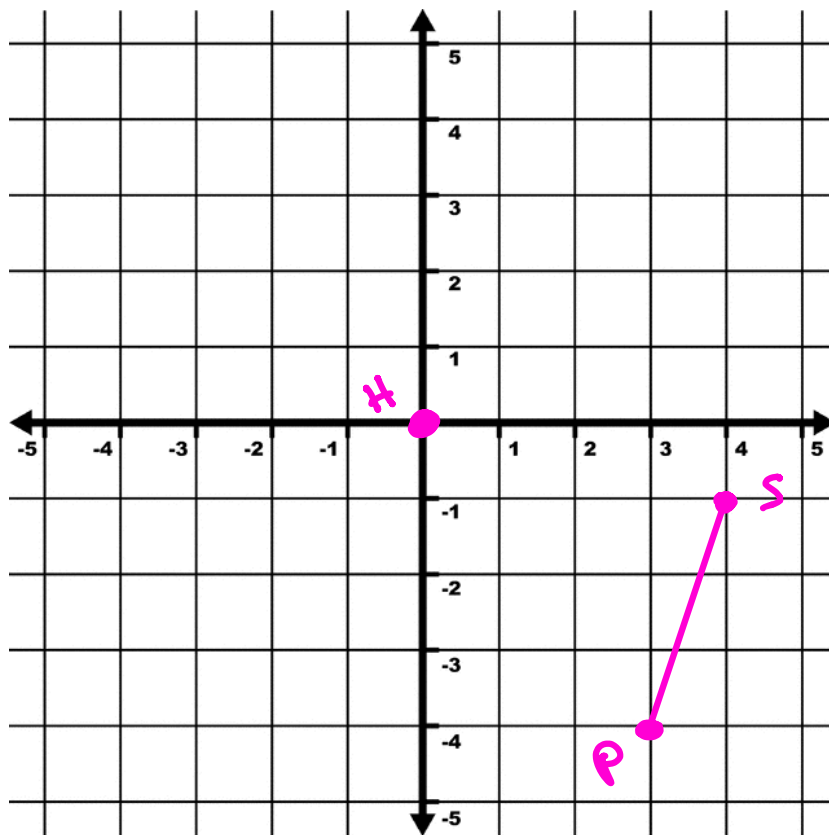
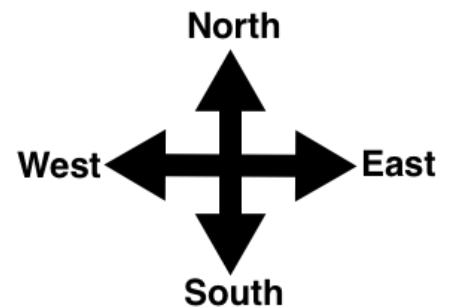
Find the distance between the two points $K(-12, 3)$ and $T(8, -16)$. Round your answer to the nearest tenth, if necessary.

$$d = \sqrt{(8 - (-12))^2 + (-16 - 3)^2}$$
$$\sqrt{761}$$

$$\approx 27.6 \text{ units}$$

Example:

Your school is 4 miles east and 1 mile south of your home. A park is 3 miles east and 4 miles south of your home. Find the distance between the park and your school.



$$S(4, -1)$$
$$P(3, -4)$$

Name

P. — #



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